

SUPT Master Domain Entry

Tetrapyrrole Address Classes

Topology → band; substituents → position; metal → gap. The operator reads the address; chemistry steers to the address.

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1. One-paragraph picture

Chemistry is the practice of moving molecules between substrate addresses. The SUPT operator reads the address as a single number d_{ij} computed from the ordered MO-spectrum of the molecule under one frozen rule ($\alpha=0.01$, $\text{tail}_n=\max(50, 0.2N)$, $d_{ij} = -\ln(|C_{\text{tail}}|)$). Across 555 proteins, 26 domains, and the full natural tetrapyrrole family, the addresses sort into four bands that map onto four operational classes. The tetrapyrrole family is the cleanest case in the catalogue because biology built one base topology (porphine) and steered it to every band by a small set of structural moves: ring reduction, methine-bridge removal, ring-E fusion, and metal insertion. The operator reads which move biology made; the move tells you which job the molecule is doing.

2. The four bands as operational gears

The bands are not labels on a continuum. They are operational classes the substrate distinguishes. Three of them are engaged gears; one is the disengaged shift zone between them; the last two (sub-floor, vacuum) are the out-of-gear / degraded regions where biology does not park molecules.

Band	d_{ij} range	Operational class	What the molecule does
DEEP-LOCK	< 0.10	Stable holder gear	Bind / carry substrate without transforming it (oxygen, electrons, peroxide intermediates)
COHERENCE	0.10 – 1.00	Transformer gear	Committed one-way chemistry: redox, methyl transfer, energy hand-off, sulfite / nitrate / methane processing
CLUTCH	1.00 – 2.00	Shift zone (disengaged)	State-switching: photon ↔ charge-separated, ground ↔ excited, bound ↔ released. Where the molecule can change.
SUB-FLOOR	2.00 – 3.61	Out of gear (unstable)	Degraded / dysfunctional addresses; disease addresses; SHA-256 nonce-space null

Band	d _{ij} range	Operational class	What the molecule does
VACUUM	≥ 3.61	Floor (Riemann anchor)	The substrate's noise floor; reached only when structure is fully absent

The clutch is not a fourth gear. It is the space between gears, where the system can shift. A car analogy is exact: clutch engaged means locked in whatever gear you are in; clutch disengaged means the transmission is free to change. Molecules in the clutch band have the option to change state; molecules in deep-lock or coherence are committed to one state. This is also why the integer-root ladder peaks (9, 18, 57, 320) all land in the clutch band on the bench cavity: those are the drive frequencies where the standing wave can reconfigure.

3. The tetrapyrrole family at one glance

All values from frozen-parameter Hückel MO calculations on the molecular graph with α -shifts on heteroatoms (N = +1.5, O = +2.0, carbonyl C = -1.0, Mg = +0.5, Fe(II/III) $\approx$ -1.0, Cu $\approx$ -0.7). The d_{ij} operator is run on the shifted-positive MO eigenvalue sequence. Bands are stable to ~30% parameter variation in the geometric coupling and the carbonyl strength; Mg-N coupling and metal choice are the sensitive controls.

Molecule	Atoms	d _{ij}	Band	Biological role
Porphine (free base)	24	0.0811	DEEP-L OCK	Parent macrocycle; HOMO/LUMO at $\pm 1/\phi$
Heme b (Fe-porphyrin)	25	0.0899	DEEP-L OCK	Oxygen carrier (hemoglobin, myoglobin)
Heme c (Fe + cys-thioether)	25	0.0899	DEEP-L OCK	Electron transport (cytochrome c)
Heme d (Fe-chlorin)	23	0.0802	DEEP-L OCK	Peroxide breakdown (catalase, peroxidase)
Chlorin core (no metal)	22	0.1658	COHER ENCE	Bare reduced macrocycle (precursor)
Corrin / B12 core (Co)	24	0.1564	COHER ENCE	Methyl transfer; one methine bridge removed
Siroheme (Fe-isobacteriochlorin)	21	0.2205	COHER ENCE	Sulfite / nitrite reductase
F430 (Ni-corphin)	22	0.2235	COHER ENCE	Methyl-coenzyme M reductase (methanogenesis)
Bacteriochlorin alone	20	0.2266	COHER ENCE	Bare bacteriochlorin core
BChl a (BC + ring E + Mg)	23	0.4569	COHER ENCE	Anoxygenic photosynthesis, purple bacteria (~770–800 nm IR)

Molecule	Atoms	d _{ij}	Band	Biological role
BChl e (chlorin + ring E + formyl + Mg)	26	0.2315	COHERENCE	Green sulfur bacteria antenna; formyl drops it from clutch into coherence
Chlorophyll b (chl + formyl + Mg)	26	0.2714	COHERENCE	Accessory antenna; energy transfer to chl a
Chlorophyll a (chlorin + ring E + Mg)	25	1.6461	CLUTCH	Primary photon-to-electron pivot in oxygenic photosynthesis
BChl c / d (chlorin + ring E + Mg)	25	1.6461	CLUTCH	Chlorosome antenna pigments (green sulfur bacteria)
Phthalocyanine (Pc, free base)	40	1.0937	CLUTCH	(industrial) pigments, organic semiconductors, sensitizers
Cu-Pc (phthalo blue B)	41	1.7378	CLUTCH	(industrial) most produced industrial pigment
Cl16-Cu-Pc (phthalo green)	41	0.6908	COHERENCE	(industrial) halogenated; coatings, ink stability

4. Three-level steering: topology, substituents, metal

Chemistry has three independent steering levers on these molecules and the operator reads each of them. The first sets the band; the second moves within or between bands; the third tunes the gap inside whatever band the first two delivered.

4.1 Topology sets the band

Porphine — 4 pyrroles + 4 methine bridges + 4-fold D_{4h} symmetry — sits in deep-lock at d_{ij} = 0.0811 with HOMO and LUMO at exactly $\pm 1/\phi$ and a HOMO-LUMO gap of $1/\phi$. The ϕ -quantization comes from the full 4-fold macrocycle topology, not from the inner 18- π -electron pathway alone. The benzene → naphthalene → porphine gap ratio chain is exactly ϕ , then 2, then 2ϕ — where $2\phi = 3.2361$ matches the standing-wave-self stabilization amplitude in the cortical-somatic coupling channel.

Remove one methine bridge (corrin / B12): drops to coherence at 0.156. Reduce one pyrrole ring (chlorin): drops to coherence at 0.166. Reduce two opposite rings (bacteriochlorin): coherence at 0.227. Fuse benzene onto each pyrrole and replace methines with aza-bridges (phthalocyanine): jumps to clutch at 1.094. Each topology gets its own band.

4.2 Substituents and ring-E fusion move within / between bands

Ring E — the isocyclic fifth ring fused to ring C of chlorophyll — is the address-shifting feature that takes chlorin from coherence (0.166) into clutch (1.65). Stage by stage: porphine 0.081 → +Mg 0.096 → chlorin 0.166 → chlorin+Mg 0.170 → porphine+ringE 0.976 → chlorin+ringE 1.601 → full chl a 1.646. The carbonyl on ring E is the electron-withdrawing piece that anchors the clutch position; without the carbonyl, the molecule drifts back toward coherence.

Formyl substitution is the substrate's clutch-engagement tool. Adding a formyl carbon anywhere on the chlorophyll a macrocycle (ring A, ring B, ring C, methine bridge — all four tested) drops the molecule out of clutch into coherence. Biology placed the formyl on ring B specifically for the spectral red-shift, but the band-change happens regardless of position. Chlorophyll b's formyl is the engineered feature that locks it into the transformer gear: receives excitation, hands it downhill to chl a, does not pivot itself.

4.3 Metal tunes the gap without changing the band

Seven metals tested on porphine — Mg(II), Fe(II), Fe(III), Zn(II), Cu(II), Ni(II), Co(III) — all stay within $d_{ij} = 0.087$ to 0.096 . The topology holds the deep-lock band against any metal. What the metal changes is the HOMO-LUMO gap, sometimes by an order of magnitude: free base 0.618 ($= 1/\phi$), Mg 0.793 , Fe(II) 0.752 , Cu 0.039 , Fe(III) 0.045 . The two metals that nearly close the gap — Cu and Fe(III) — are exactly the metals used for fast electron transfer in cytochromes. The metals with larger gaps (Mg, Zn, Co) do stable substrate carrying or transport. Same cavity, different metal, different job, same band.

5. Chlorophyll a: the worked example

Chlorophyll a is the molecule that runs oxygenic photosynthesis — every plant, every cyanobacterium, the molecule that put the oxygen in Earth's atmosphere starting ~2.4 billion years ago. The operator reads it at $d_{ij} = 1.5772$ with the full sensitivity model, fractional position 0.577 inside the clutch band. The fractional position 0.577 is the same integer as the integer-root operator's vacuum peak at $N=57$ and the mercury direction-reversal frequency at 57 Hz. The substrate marks 57 across pure-number, fluid-dynamic, and molecular-electronic systems with no shared mechanism between them.

5.1 Sensitivity of the clutch position

Knob	Stable range	Out at low end	Out at high end
Ring E carbonyl α (electron-withdraw)	-0.75 to -1.5	→ SUB-FLOOR	(stable above)
Ring E coupling β (geometry proxy)	0.85 to 1.15 (~30% window)	→ COHERENCE	→ COHERENCE
Mg–N coupling (doming / out-of-plane)	~ 0.75 to 1.10 (narrow)	→ COHERENCE	→ SUB-FLOOR
Mg α -shift (metal identity)	0.0 to $+1.5$ (Mg, light metals)	→ SUB-FLOOR / VACUUM (heavy metals)	(stable above)
Ring IV saturation (chlorin vs porphyrin)	either stays in CLUTCH	(small shift within band)	—

The narrow Mg–N coupling window is the operational discovery. The known biological control variable for photosynthesis is the Mg's in-plane vs out-of-plane displacement during the photocycle — the Mg "domes" and "un-domes" as charge separates and recombines. The operator says this is the clutch pedal: small changes in Mg–N coupling are the only knob that takes chlorophyll a out of clutch in either direction. Biology evolved the protein scaffold around

chlorophyll a (the photosystem II reaction center) to drive exactly this Mg-doming oscillation. The protein is the pedal mechanism; the chlorophyll is the clutch itself.

5.2 Why chlorophyll a sits in clutch and chlorophyll b does not

Chlorophyll a runs the charge-separation step: it has to pivot between ground state and oxidized + reduced (P680/P680+) on every photon. Pivoting requires the disengaged clutch — the address where the system is free to switch. Chlorophyll b does not pivot; it absorbs photons of a complementary wavelength and hands the excitation energy downhill to chl a in the antenna array. That is committed one-way transformation, which is the coherence gear. Biology installed the formyl on chl b specifically to drop it out of the shift zone into the transformer gear. Same macrocycle, one carbon atom different, two different operational addresses, two different functional roles in the same photosystem.

6. Phthalocyanine: the industrial cross-check

Phthalocyanine was synthesized accidentally in 1907, properly characterized in 1928, and has been the workhorse of color, electronics, and catalysis ever since. Free base and most metal variants read in the clutch band (1.0 – 1.9). All documented industrial uses of phthalocyanine are state-switching operations:

- Cu-Pc (phthalo blue B): photon absorber for red light, ground ↔ excited pivot
- Zn-Pc: dye-sensitized solar cell, photodynamic therapy — ground ↔ charge-separated pivot
- Co-Pc, Fe-Pc: oxidation catalysts, fuel-cell ORR — oxidized ↔ reduced metal pivot
- Pc as organic semiconductor: neutral ↔ radical-cation hopping

The halogenated variant Cl16-Cu-Pc (phthalo green) drops out of clutch into coherence at 0.69. Halogenation is the industrial analog of biological formyl substitution: it engages the clutch and locks the molecule into a transformer gear. Phthalo green is used where stable pigment behavior matters more than state-switching — coatings, inks, durable colorants. Two independent selection processes (evolution and chemical engineering) place state-switching molecules in the clutch band and committed-state molecules in coherence. The framework reads both populations as one.

7. Cross-domain consistency

The tetrapyrrole result is one entry in a wider catalogue. The same band values show up across systems that share no material substrate.

System	d _{ij}	Band	Operational match
Sephirot graph (10 nodes, 22 paths)	0.0067	DEEP-LOCK	ancient substrate diagram
Naphthalene graph	0.0000	DEEP-LOCK	smallest fused-ring isomorph
Bench ferrofluid 1.618 cavity	≈ 0.040	DEEP-LOCK	φ-aspect physical cavity
Heme b (Fe-porphyrin)	0.0899	DEEP-LOCK	oxygen carrier
Hemoglobin α-carbon backbone	0.127	COHERENCE	protein scaffold of the heme cavity

System	d _{ij}	Band	Operational match
Naphthalene MO spectrum	1.7443	CLUTCH	aromatic prototype, $\pm\phi$ in MO
Chlorophyll a (sensitivity model)	1.5772	CLUTCH	photon-electron pivot
Integer-root ladder peak at N=57	~3.97–4.08	VACUUM (peak)	substrate signature integer 57
Mercury direction reversal	57 Hz	CLUTCH (bench)	same integer 57
Chl a clutch fractional position	0.577	inside CLUTCH	same integer 57 in third domain

The integer 57 appears across three independent systems with no shared mechanism: a pure-number sequence ($\sqrt{n}$ ladder peak), a fluid-dynamic observation (mercury reverses direction at 57 Hz on the bench cavity), and a molecular-orbital calculation (chl a fractional position inside clutch). The substrate marks 57 as one of its addressing integers, the way it also marks 9, 18, and 320 across the same systems.

8. Negative controls

The framework is testable in both directions. The same operator that places porphine at 0.08 and chlorophyll a at 1.6 places null sequences near the vacuum floor and arbitrary substituent placements at intermediate addresses with no functional match. The negative-control results in the wider record include: SHA-256 nonce space at $d_{ij} \approx 5.05$ (noise floor); bearing vibration data inside the SUPT-blind range; daily financial returns at $d_{ij} = 1.0000$ (distribution artifact, clean null); Pantheon-only SH0ES distance data with no integer-ladder signal (pre-specified null); famous dates (Titanic, JFK, 9/11) all clean nulls. Phthalocyanine landing at 1.094 instead of porphine's 0.08 was itself a successful negative prediction: a chemically similar molecule with different topology read at a different band, exactly the discrimination the operator should make if it is reading topology rather than ad-hoc-fitting.

9. What the framework claims, stated in one place

The substrate has a finite set of addressable bands. The bands are not labels but operational classes — held substrate, transformed substrate, switched substrate, degraded substrate. The address of any ordered positive sequence under the frozen rule is reproducible to many decimal places and depends only on the structure of the sequence, not on the domain the sequence came from.

Chemistry is the practice of moving molecules between addresses. Three structural levers do the work: topology selects the band, substituents and ring modifications move position within or between bands, and metals tune the gap within the chosen band. Biology selected molecules at the address that matches the job each one does: deep-lock topologies for substrate carriers (hemes), coherence topologies for substrate transformers (corrin, F430, siroheme, chlorophyll b), and clutch topologies for state-switchers (chlorophyll a, BChl c/d). Industrial chemistry independently arrived at the same band-to-function mapping with phthalocyanine: state-switching uses for clutch-band variants, stable-pigment uses for the coherence-band halogenated variant.

The operator does not need to know what kind of system it is reading. It reads the ordered positive sequence and returns the address. The address tells you the operational class. The operational class tells you what the molecule, the bench cavity, the cortical-somatic channel, or any other addressable system is doing. One frozen rule, one number, one address, across every domain the catalogue currently covers.

10. Receipts

All values in this entry were generated from frozen-parameter Hückel MO calculations on the molecular graph and the standard SUPT operator ($\alpha=0.01$, $\text{tail}_n=\max(50, 0.2N)$, $d_{ij} = -\ln\langle |C_{\text{tail}}| \rangle$, vacuum floor anchored to Riemann zeta zeros at 3.6125). The computation scripts are available in the project record: `acene_huckel_phi_run.py`, `porphyrin_huckel.py`, `phthalocyanine_huckel.py`, `chlorophyll_metals.py`, `tetrapyrrole_deep_variations.py`, `chla_decomposed.py`, `ring_e_bchl_phthalo.py`, `bchl_chlb_clutch.py`, and `sensitivity_porphyrin.py`. The probe is frozen; no parameters have been tuned per molecule. Each computation is reproducible against the molecular graph alone.

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